

Extension of kinetic theory to cohesive particle flow

Hae Kim¹, Hamid Arastoopour^{*}

Department of Chemical and Environmental Engineering, Illinois Institute of Technology, 10 W. 33rd Street, Chicago, IL 60616, USA

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Abstract

Particle flow is significantly affected by the agglomeration of particles due to the inter-particle force when fine particles, such as Geldart's Group C particles, are used. In order to quantify cohesive particle flow behavior, governing equations based on the kinetic theory for cohesive particle flow were developed. In the derivation of the governing equations for cohesive particle flow, we defined a new distribution function of the instantaneous velocity of the particle relative to the average velocity based on the volume fraction of the particulate phase that is conserved upon agglomeration. Furthermore, to account for the effect of diameter growth on agglomeration, we considered the change in number of particles due to agglomeration in the derivation of a new conservation of the number of particles equation. Based on our distribution function, governing equations were derived; namely, mass and momentum plus energy and conservation of number of particles equations for cohesive particle flow. This set of equations is capable of describing cohesive particle flow behavior as well as particle diameter variation due to agglomeration. Finally, to show the validity of our model, we analyzed the homogeneous simple shear flow of particles under the agglomeration condition. The predicted flow properties, such as shear viscosity, normal viscosity, and particle growth, agreed well with the expected trends. © 2002 Elsevier Science B.V. All rights reserved.

Keywords: Kinetic theory; Cohesive particle; Agglomeration; Gas/solid flow; Granular flow

1. Introduction

The dynamics of inter-particle collisions in non-cohesive particle flow systems have been reconsidered based on the concept of the kinetic theory [4], and further modified and applied to the particulate phase by Jenkins and Savage [6], Savage and Jeffrey [14], Savage [15], Savage and Sayed [16], Lun et al. [11], Johnson and Jackson [9], Jenkins and Richman [7,8], Ding and Gidaspow [5], Sinclair and Jackson [17], Louge et al. [10], and Arastoopour and Kim [2]. The simulations of gas/non-cohesive particle flow systems using the kinetic theory model by Ding and Gidaspow [5] and Sinclair and Jackson [17], and the experimental verification of the Maxwellian type distribution of cohesive solid particles by Zhang et al. [19] showed that the adoption of the gas kinetic theory for gas/particle systems is a promising approach. Further-

more, experimental data on particle fluctuation velocity using a laser doppler anemometer (Yang and Arastoopour [1] and Zhang and Arastoopour [18]) showed the significance of the fluctuation velocity in identifying the particulate phase flow patterns in gas/agglomerating particle flow systems. This also revealed the need for a model such as the kinetic theory that considers fluctuation of the particle as a major parameter. However, in the literature, models developed for non-cohesive particles do not have the capability of describing flow behavior for cohesive particle systems because the most significant characteristic of cohesive particle flow is the formation of agglomerates, which considerably affect flow patterns. The formation of agglomerates results in a reduction in the number and an increase in the size of particles, both of which directly affect the inter-particle collisions and, in turn, the particle phase properties such as viscosity and pressure, as well as gas/particle drag force in gas/particle flow systems. Because the number of particles is not conserved upon agglomeration, the Maxwellian–Boltzmann integral–differential equation needs to be evaluated each time particle agglomeration occurs. To overcome this problem, we defined a new distribution function of the instantaneous velocity of the particles relative to the average velocity based on the volume fraction of the particulate phase,

^{*} Corresponding author. Tel.: +1-312-567-3040; fax: +1-312-567-8874.

E-mail address: arastoopour@iit.edu (H. Arastoopour).

¹ Current address: Department of Robot System Engineering, Tongmyong University of Information Technology, Namgu, Pusan 608-080, Republic of Korea.

which is conserved upon agglomeration. Furthermore, to account for the effect of diameter growth on agglomeration, we considered a new conservation of the number of particles equation. Based on our distribution function, governing equations were derived; namely, conservation of mass, momentum, fluctuation energy, and number of particles equations. This set of equations is capable of describing particle flow behavior as well as the particle diameter variation due to agglomeration.

2. Agglomeration process

The agglomeration of two particles upon collision is a phenomenon which is directly related to surface forces and interaction of boundaries of two particles. In the agglomeration process, not only is the kinetic energy of the colliding particles dissipated or stored, but also significant surface cohesion forces result in the binding of particles. Clustering may also enhance the agglomeration process. Clustering is a phenomenon, which is associated with energy dissipation due to the collision of particles irrespective of the particle surface forces. Clustering affects the particle flow patterns and, in turn, the particle flow properties.

In this study, we focused on the formation of binary agglomerates due to surface forces, irrespective of whether the agglomeration process is enhanced by the formation of clusters. In binary collisions of two agglomerating particles, unlike non-agglomerating particles, the velocities of the two particles after collision become equal and the particle size changes. To incorporate this fundamental change in kinetic theory formulation, we first adopted the collisional model for non-cohesive particle flow proposed by Jenkins and Savage [6]. The energy loss due to the binary collision of two identical spherical particles may be written following [6] as

$$\Delta E = \frac{m}{4} (\vec{c}_{12}^2 - \vec{c}_{12}'^2) \quad (1)$$

where, $\vec{c}_{12}$ and $\vec{c}_{12}'$ are the relative velocity of the two particles before and after collision. Then, we consider that the inelasticity of the particles has an effect not only on the direction of the collision ($\vec{k}$) but also on the other orthogonal directions ($\vec{l}, \vec{m}$). The relative velocity variation and energy loss due to the inelastic collision may be written as

$$\begin{aligned} (\vec{c}_{12}' \cdot \vec{k}) &= -e(\vec{c}_{12} \cdot \vec{k}) \\ (\vec{c}_{12}' \cdot \vec{j}) &= f(\vec{c}_{12} \cdot \vec{j}) \quad \text{for } \vec{j} = \vec{l}, \vec{m} \end{aligned} \quad (2)$$

$$\Delta E_{\text{inelastic}} = \frac{m}{4} \left[(f^2 - e^2)(\vec{c}_{12} \cdot \vec{k}) + (1 - f^2)\vec{c}_{12}^2 \right] \quad (3)$$

where, e is the normal restitution coefficient, and f is the tangential roughness coefficient.

Agglomeration is a result of the interaction of the surface energy of two particles. Let E_c be the contact bonding energy, which is a function of the surface property of the material. The total energy loss when cohesive particles are colliding, $\Delta E_{\text{cohesive}}$, is equal to $(\Delta E_{\text{inelastic}} + E_c)$

$$\begin{aligned} \Delta E_{\text{cohesive}} &= \frac{m}{4} \left[(f^2 - e^2)(\vec{c}_{12} \cdot \vec{k})^2 \right. \\ &\quad \left. + \left(1 - f^2 + \frac{4E_c}{m\vec{c}_{12}^2} \right) \vec{c}_{12}^2 \right]. \end{aligned} \quad (4)$$

From Eqs. (1) and (4), the relationship between relative velocity before and after collision is obtained as

$$\begin{aligned} (\vec{c}_{12}' \cdot \vec{k}) &= -\sqrt{e^2 - \frac{4E_c}{m\vec{c}_{12}^2}} (\vec{c}_{12} \cdot \vec{k}) \\ (\vec{c}_{12}' \cdot \vec{j}) &= \sqrt{f^2 - \frac{4E_c}{m\vec{c}_{12}^2}} (\vec{c}_{12} \cdot \vec{j}) \quad \text{for } \vec{j} = \vec{l}, \vec{m} \end{aligned} \quad (5)$$

and

$$\begin{aligned} \vec{c}_{12}' - \vec{c}_{12} &= - \left[\left(1 - \sqrt{f^2 - \frac{4E_c}{m\vec{c}_{12}^2}} \right) \vec{c}_{12} \right. \\ &\quad \left. + \left(\sqrt{e^2 - \frac{4E_c}{m\vec{c}_{12}^2}} \right. \right. \\ &\quad \left. \left. + \sqrt{f^2 - \frac{4E_c}{m\vec{c}_{12}^2}} \right) (\vec{c}_{12} \cdot \vec{k}) \vec{k} \right]. \end{aligned} \quad (6)$$

Using Eqs. (5) and (6), the velocity of the cohesive particles after collision becomes

$$\begin{aligned} \vec{c}_1' &= \vec{c}_1 - \frac{1}{2} \left[\left(1 - \sqrt{f^2 - \frac{4E_c}{m\vec{c}_{12}^2}} \right) \vec{c}_{12} \right. \\ &\quad \left. + \left(\sqrt{e^2 - \frac{4E_c}{m\vec{c}_{12}^2}} + \sqrt{f^2 - \frac{4E_c}{m\vec{c}_{12}^2}} \right) (\vec{c}_{12} \cdot \vec{k}) \vec{k} \right] \end{aligned} \quad (7)$$

$$\begin{aligned} \vec{c}_2' &= \vec{c}_2 + \frac{1}{2} \left[\left(1 - \sqrt{f^2 - \frac{4E_c}{m\vec{c}_{12}^2}} \right) \vec{c}_{12} \right. \\ &\quad \left. + \left(\sqrt{e^2 - \frac{4E_c}{m\vec{c}_{12}^2}} + \sqrt{f^2 - \frac{4E_c}{m\vec{c}_{12}^2}} \right) (\vec{c}_{12} \cdot \vec{k}) \vec{k} \right]. \end{aligned} \quad (8)$$

Considering Eqs. (1) and (4), we may rewrite two equations as

$$\begin{aligned}\vec{c}_{12}'^2 &= (e^2 - f^2)(\vec{c}_{12} \cdot \vec{k})^2 + \left(f^2 - \frac{4E_c}{m\vec{c}_{12}^2}\right)\vec{c}_{12}^2 \\ &= \left(e^2 - \frac{4E_c}{m\vec{c}_{12}^2}\right)(\vec{c}_{12} \cdot \vec{k})^2 \\ &\quad + \sum_{j=1}^2 \left(f^2 - \frac{4E_c}{m\vec{c}_{12}^2}\right)(\vec{c}_{12} \cdot \vec{j})^2.\end{aligned}\quad (9)$$

For the case of agglomeration, $\vec{c}_1' = \vec{c}_2'$ and $\vec{c}_{12}' = 0$. Thus the general criterion of agglomeration may be written as

$$e^2 \leq \frac{4E_c}{m\vec{c}_{12}^2} \text{ and } f^2 \leq \frac{4E_c}{m\vec{c}_{12}^2}. \quad (10)$$

On the other hand, Lun [12] studied the effect of the roughness coefficient in the tangential direction on normal and shear stress, compared the results with experimental data, and showed good agreement using the value of the roughness coefficient as -1 ; -0.8 and -0.5 at the value of the restitution coefficient in the normal direction as 0.8 . This and Eq. (10) imply that the appropriate value for the tangential roughness coefficient for cohesive particles, in terms of the effective normal restitution coefficient, may be approximately equal to the negative value of the normal restitution coefficient when no spin or spin reversal occurs following the collision.

Introducing new dimensionless parameter λ_c as

$$\lambda_c = \frac{4E_c}{me^2\vec{c}_{12}^2} \quad (11)$$

the relationship between relative velocity before and after collision may be obtained as

$$\begin{aligned}(\vec{c}_{12}' \cdot \vec{k}) &= -e\sqrt{1 - \lambda_c}(\vec{c}_{12} \cdot \vec{k}) \\ (\vec{c}_{12}' \cdot \vec{j}) &= e\sqrt{1 - \lambda_c}(\vec{c}_{12} \cdot \vec{j}) \quad \text{for } \vec{j} = \vec{l}, \vec{m}\end{aligned}\quad (12)$$

and

$$\vec{c}_{12}' - \vec{c}_{12} = -\left[(1 - e\sqrt{1 - \lambda_c})\vec{c}_{12} + 2e\sqrt{1 - \lambda_c}(\vec{c}_{12} \cdot \vec{k})\vec{k}\right]. \quad (13)$$

Eq. (12) means, for cohesive particles, the effective restitution coefficient $e\sqrt{1 - \lambda_c}$ is a function of the contact bonding energy as well as the relative velocity of the particles before collision.

Using Eqs. (12) and (13), the velocity of the cohesive particles after collision upon particle separation becomes

$$\begin{aligned}\vec{c}_1' &= \vec{c}_1 - \frac{1}{2}\left[(1 - e\sqrt{1 - \lambda_c})\vec{c}_{12} + 2e\sqrt{1 - \lambda_c}(\vec{c}_{12} \cdot \vec{k})\vec{k}\right] \\ \vec{c}_2' &= \vec{c}_2 + \frac{1}{2}\left[(1 - e\sqrt{1 - \lambda_c})\vec{c}_{12} + 2e\sqrt{1 - \lambda_c}(\vec{c}_{12} \cdot \vec{k})\vec{k}\right]\end{aligned}\quad (14)$$

and the velocity of cohesive particles after agglomeration, when the effective restitution coefficient $e\sqrt{1 - \lambda_c}$ is zero in all directions, becomes

$$\begin{aligned}\vec{c}_1' &= \vec{c}_1 - \frac{1}{2}\vec{c}_{12} \\ \vec{c}_2' &= \vec{c}_2 + \frac{1}{2}\vec{c}_{12}\end{aligned}\quad (15)$$

thus, $\vec{c}_1' = \vec{c}_2'$.

3. Conservation equations

Due to agglomeration, the number of particles per unit volume is no longer conserved, so the Maxwell-Boltzmann integral-differential equation needs to be evaluated at each agglomeration occurrence. In order to overcome this problem and to modify the distribution function for particles with finite size, we defined a new distribution function of the instantaneous particle velocity based on the volume fraction of the particulate phase which was conserved throughout the agglomeration with no phase change. The probable volume of particles which, at time t , are situated in the volume element $\vec{r}$, $d\vec{r}$ and have a velocity in the range $\vec{c}$, $d\vec{c}$, is equal to $f(\vec{c}, \vec{r}, t)d\vec{c}d\vec{r}$. The whole volume of particles in the element $d\vec{r}$ is obtained by integrating $f(\vec{c}, \vec{r}, t)d\vec{c}d\vec{r}$ throughout the whole velocity space. This whole volume is $\varepsilon_s d\vec{r}$. Then we can write ε_s as

$$\varepsilon_s = \int_{\vec{c}} f(\vec{c}, \vec{r}, t)d\vec{c} \quad (16)$$

then the mean value of the function ψ which is averaged over the particles with all possible velocities is

$$\langle \psi \rangle = \frac{1}{\varepsilon_s} \int_{\vec{c}} \psi f(\vec{c}, \vec{r}, t)d\vec{c}. \quad (17)$$

Although it is rather impossible to have a perfect collision in particle flow systems, we assumed that our volume distribution function of the instantaneous velocities of the particles is the Maxwellian type. The validity of this assumption for both cohesive and non-cohesive particles was confirmed experimentally by Arastoopour and Yang [1] and Zhang et al. [19]. We define f_1 and f_2 as the distribution functions for particles 1 and 2 at time t , point of contact $\vec{r}$, and the instantaneous velocities of $\vec{c}_1$ and $\vec{c}_2$, respectively. Thus, for example,

$$f_1 = \frac{\varepsilon_s}{(2\pi\Theta)^{\frac{3}{2}}} \exp\left[-\frac{(\vec{c}_1 - \vec{U}_s)^2}{2\Theta}\right] \quad (18)$$

where Θ is the fluctuation temperature and $\vec{U}_s$ is the mean velocity of the particle.

Agglomeration results in the variation of the number of particles (i.e., change of particle size). Consider collision between the pairs of particles 1 and 2, when $\lambda_c \geq 1$ agglomeration occurs. Let us define V_c as

$$V_c = \sqrt{\frac{4E_c}{me^2}}. \quad (19)$$

Then the agglomeration domain is $0 < c_{12} < V_c$. We assumed an average particle size within each control volume/computational cell. Thus, collision of the particles was expressed in terms of the collision of the averaged particles in each control volume (i.e., the particle size growth is not a function of instantaneous velocity). Applying the agglomeration domain, the total number of agglomerations per unit volume and time at specific $\varepsilon_s, \vec{U}_s, \Theta, \dots$ may be written as

$$N_a = \int \int_{[0 < c_{12} < V_c]} \int_{[0 < \theta_k < \frac{\pi}{2}]} g_o f_1 f_2 \times \left[1 + \frac{1}{2} D_p \vec{k} \cdot \nabla \ln \frac{f_2}{f_1} \right] \frac{D_p^2}{V_p^2} (\vec{C}_{12} \cdot \vec{k}) d\vec{k} dc_1 dc_2. \quad (20)$$

Since N_a is assumed to be only a function of the macroscopic properties, the macroscopic balance for the conservation of the number of particles may be expressed as

$$\frac{\partial n}{\partial t} + \nabla \cdot (n \vec{U}_s) = -N_a. \quad (21)$$

We derived the following conservation of mass, momentum, and fluctuation energy equations by substituting ψ with ρ_s , $\rho_s \vec{c}$, and $\frac{1}{2} \rho_s \vec{c}^2$ into the Maxwell–Boltzmann integral–differential equation, and solving for the collisional integral using criterion of agglomeration.

4. Governing equations and constitutive equations

The following are the conservation of mass, momentum, fluctuating energy, and number of particles equations, and the constitutive equations for cohesive particle systems.

Conservation of mass

$$\frac{\partial(\varepsilon_s \rho_s)}{\partial t} + \nabla \cdot (\varepsilon_s \rho_s \vec{U}_s) = 0. \quad (22)$$

Conservation of momentum

$$\frac{\partial(\varepsilon_s \rho_s \vec{U}_s)}{\partial t} + \nabla \cdot (\varepsilon_s \rho_s \vec{U}_s \vec{U}_s) = -\varepsilon_s \rho_s \vec{g} + \nabla \cdot \mathbf{T}_s. \quad (23)$$

Conservation of fluctuation energy

$$\frac{2}{3} \left[\frac{\partial(\varepsilon_s \rho_s \Theta)}{\partial t} + \nabla \cdot (\varepsilon_s \rho_s \vec{U}_s \Theta) \right] = \mathbf{T}_s : \nabla \vec{U}_s + \nabla \cdot \vec{q}_c + \gamma. \quad (24)$$

Conservation of number of particles

$$\frac{\partial(\varepsilon_s \chi)}{\partial t} + \nabla \cdot (\varepsilon_s \vec{U}_s \chi) = -V_{po} N_a. \quad (25)$$

Constitutive Equations

Solid phase stress

$$\mathbf{T}_s = \left[-(\rho_s^k + \rho_s^c) + \varepsilon_s \mu_b (\nabla \cdot \vec{U}_s) \right] \mathbf{I} + 2 \varepsilon_s \mu_s \mathbf{S} + \mathbf{T}_f \quad (26)$$

where,

Deformation rate

$$\mathbf{S} = \frac{1}{2} \left[(\nabla \vec{U}_s) + (\nabla \vec{U}_s)^T \right] - \frac{1}{3} (\nabla \cdot \vec{U}_s) \mathbf{I}. \quad (27)$$

Frictional stress due to non-head-on collisions of cohesive particles

$$\mathbf{T}_f = \varepsilon_s \mu_f (\nabla \vec{U}_s). \quad (28)$$

Kinetic solid phase pressure

$$p_s^k = \varepsilon_s \rho_s \Theta. \quad (29)$$

Collisional solid phase pressure

$$p_s^c = 2 \varepsilon_s^2 \rho_s g_o \Theta [1 + e \xi_{p1}(R_c)]. \quad (30)$$

Solid phase bulk viscosity

$$\mu_b = \frac{2}{3\sqrt{\pi}} \varepsilon_s \rho_s g_o D_{po} \sqrt{\Theta} \chi^{-\frac{1}{3}} [1 + 3e \xi_{p2}(R_c)]. \quad (31)$$

Solid phase shear viscosity

$$\mu_s = \frac{2}{5\sqrt{\pi}} \varepsilon_s \rho_s g_o D_{po} \sqrt{\Theta} \chi^{-\frac{1}{3}} [1 + 3e \xi_{p2}(R_c)]. \quad (32)$$

Solid phase frictional viscosity

$$\mu_f = \frac{2}{\sqrt{\pi}} \varepsilon_s \rho_s g_o D_{po} \sqrt{\Theta} \chi^{-\frac{1}{3}} [1 - e \xi_{p2}(R_c)]. \quad (33)$$

Radial distribution function following Carnahan and Starling [3]

$$g_o = \frac{3}{5} \left[1 - \left(\frac{\varepsilon_s}{\varepsilon_m} \right)^{\frac{1}{3}} \right]^{-1}. \quad (34)$$

Cohesive pressure coefficient

$$\xi_{p1}(R_c) = \exp(-R_c^2) \left(1 + \frac{2}{3} R_c^2\right). \quad (35)$$

Cohesive viscosity coefficient

$$\begin{aligned} \xi_{p2}(R_c) &= \exp(-R_c^2) \left(1 + \frac{3}{4} R_c^2 + \frac{3}{16} R_c^4\right) - \frac{1}{32} R_c^6 E_i(R_c^2) \\ &\quad - \frac{1}{32} R_c^6 \sum_{j=1}^{\infty} \frac{(-1)^j \prod_{k=1}^{j+1} (2k+1)}{2^j j! (j+3)!} \left[R_c^{2j} E_i(R_c^2) \right. \\ &\quad \left. + \exp(-R_c^2) \sum_{n=1}^j (-1)^n (n-1)! R_c^{2(j-n)} \right]. \quad (36) \end{aligned}$$

Dimensionless ratio of contacting bonding energy to fluctuation energy

$$R_c = \sqrt{\frac{6E_c \chi}{\pi \rho_s e^2 D_{po}^3 \Theta}}. \quad (37)$$

Dimensionless ratio of initial volume of the particle to the volume of the particle after agglomeration

$$\chi = \frac{D_{po}^3}{D_p^3}. \quad (38)$$

Flux of fluctuation energy

$$\vec{q}_c = k_s \nabla \Theta \quad (39)$$

where, Conductivity of fluctuation energy

$$k_s = \frac{4}{\sqrt{\pi}} \varepsilon_s^2 \rho_s g_o D_{po} \chi^{-\frac{1}{3}} \sqrt{\Theta}. \quad (40)$$

Collisional energy dissipation

$$\begin{aligned} \gamma &= -\varepsilon_s^2 \rho_s g_o \Theta \left\{ \frac{24\chi^{\frac{1}{3}}}{D_{po}} \sqrt{\frac{\Theta}{\pi}} [1 - e^{\xi_{\gamma 1}(R_c)}] \right. \\ &\quad \left. - 5(\nabla \cdot \vec{U}_s) [1 - e^{\xi_{\gamma 2}(R_c)}] \right\} \quad (41) \end{aligned}$$

where, Cohesive dissipation coefficient

$$\xi_{\gamma 1}(R_c) = \exp(-R_c^2) \left(1 + \frac{1}{2} R_c^2\right) \quad (42)$$

$$\begin{aligned} \xi_{\gamma 2}(R_c) &= \operatorname{erfc}(R_c) \left(1 - \frac{2}{5} R_c^2\right) \\ &\quad + \exp(-R_c^2) \left(\frac{2}{\sqrt{\pi}} R_c + \frac{8}{15\sqrt{\pi}} R_c^3\right). \quad (43) \end{aligned}$$

Number of agglomerations per unit volume and time

$$\begin{aligned} N_a &= \frac{4\chi \varepsilon_s^2 g_o}{V_{po}} \left\{ \frac{6\chi^{\frac{1}{3}}}{D_{po}} \sqrt{\frac{\Theta}{\pi}} [1 - \xi_{a1}(R_c)] \right. \\ &\quad \left. - (\nabla \cdot \vec{U}_s) [1 - \xi_{a2}(R_c)] \right\} \quad (44) \end{aligned}$$

where, Agglomeration number coefficient

$$\xi_{a1}(R_c) = \exp(-R_c^2) (1 + R_c^2) \quad (45)$$

$$\xi_{a2}(R_c) = \operatorname{erfc}(R_c) + \exp(-R_c^2) \left(\frac{2}{\sqrt{\pi}} R_c + \frac{4}{3\sqrt{\pi}} R_c^3\right). \quad (46)$$

5. Shear flow analysis

In this section, we focus our attention on the case of a homogeneous simple shear flow with no spatial gradients of solid volume fraction, fluctuation velocity, and particle diameter under agglomerating conditions. The present shear flow analysis is based upon the assumption that the rate of energy dissipation is in the moderate range; thus, the

dimensionless parameter $R_t = \left(\left| \frac{dU_{s,x}}{dy} \right| \right) / \langle \bar{C}^2 \rangle^{\frac{1}{2}}$, defined by Savage and Jeffrey [14] as the ratio of the characteristic mean shear velocity to the r.m.s. of the instantaneous particle fluctuation velocity is in the intermediate range of the order one. Savage and Jeffrey [14] showed good agreement between their theoretical prediction for R_t and the experimental data at $R_t = 1.7$, and Lun [12] also reported that R_t can be of the order one when the collision becomes relatively less elastic. On the other hand, we may rewrite the dimensionless parameter R_c , defined in the previous section as the ratio of the contacting bonding energy to the fluctuation energy, in terms of R_t as

$$R_c = R_t \sqrt{\frac{18E_c \chi^{\frac{5}{3}}}{\pi \rho_s e^2 D_{po}^5 \left| \frac{dU_{s,x}}{dy} \right|^2}}. \quad (47)$$

One of the limitations of our present model is that R_c should be of the same or less order as R_t , which implies that R_c is in the intermediate range of the order one. The method used for our homogeneous simple shear flow analysis is similar to that of Richman and Chou [13], who derived the kinetic theory for homogeneous shear flow of identical spheres that undergo gradual size reduction due

to the cumulative effects of particle wear caused by repeated binary collisions. Our kinetic theory is developed based on the total mass conservation of the system, but the Richman and Chou theory is developed based on the total mass loss of the system by neglecting the small fraction of the particle, which may remain in the system. In this section, we would like to examine the capability of our theory in predicting agglomeration and fluctuation velocity in the case of a homogeneous simple shear flow. In a homogeneous simple shear flow, we may write

$$\varepsilon_s \neq f(x \text{ and } y) \quad (48)$$

$$U_{s,x} = f(y) \quad (49)$$

$$U_{s,y} = 0 \quad (50)$$

then, the conservation of mass becomes

$$\frac{d(\varepsilon_s \rho_s)}{dt} = 0. \quad (51)$$

This results in constant value of ε_s .

For a constant shear rate of $\left| \frac{dU_{s,x}}{dy} \right|$, the shear stress becomes

$$T_{s,yx} = \varepsilon_s (\mu_s + \mu_f) \left| \frac{dU_{s,x}}{dy} \right|. \quad (52)$$

Assuming Θ and χ to be only functions of time, the conservation of momentum becomes

$$\frac{dT_{s,yx}}{dy} = 0, \quad (53)$$

the conservation of fluctuation energy becomes

$$\frac{2}{3} \varepsilon_s \rho_s \frac{d\Theta}{dt} = T_{s,yx} : \left| \frac{dU_{s,x}}{dy} \right| + \gamma \quad (54)$$

and the conservation of number of particles may be written as

$$\varepsilon_s \frac{d\chi}{dt} = -V_{po} N_a. \quad (55)$$

By substituting the constitutive equations in the previous section into Eqs. (54) and (55), and introducing t^* = $\left| \frac{dU_{s,x}}{dy} \right| t$ as dimensionless time and using the dimensionless variable R_c , which is the ratio of contacting bonding energy to fluctuating energy, we may have

sionless variable R_c , which is the ratio of contacting bonding energy to fluctuating energy, we may have

$$\frac{d\chi}{dt^*} = - \frac{12\sqrt{6} \varepsilon_s g_o \sqrt{E_c}}{\pi e \sqrt{\rho_s} D_{po}^2 \left| \frac{dU_{s,x}}{dy} \right|} \frac{\chi^{\frac{11}{6}}}{R_c} [1 - \xi_{a1}(R_c)] \quad (56)$$

$$\begin{aligned} \frac{dR_c}{dt^*} = & - \frac{3e \varepsilon_s g_o \sqrt{\rho_s} D_{po}^2 \left| \frac{dU_{s,x}}{dy} \right|}{10\sqrt{6} \sqrt{E_c}} \frac{R_c^2}{\chi^{\frac{5}{6}}} [3 - e \xi_{p2}(R_c)] \\ & + \frac{9\sqrt{6} \varepsilon_s g_o \sqrt{E_c}}{\pi e \sqrt{\rho_s} D_{po}^2 \left| \frac{dU_{s,x}}{dy} \right|} \chi^{\frac{5}{6}} [1 - e^2 \xi_{\gamma 1}(R_c)] \\ & - \frac{6\sqrt{6} \varepsilon_s g_o \sqrt{E_c}}{\pi e \sqrt{\rho_s} D_{po}^2 \left| \frac{dU_{s,x}}{dy} \right|} \chi^{\frac{5}{6}} [1 - \xi_{a1}(R_c)]. \end{aligned} \quad (57)$$

In order to know the behavior of χ , R_c , and R_t with dimensionless time, Eqs. (56) and (57) were solved simultaneously using the fourth order Runge–Kutta numerical technique.

Fig. 1 shows the time behavior of the variables χ , R_c , and R_t . Initially, the agglomeration of particles is enhanced due to the presence of small particles; thus, the

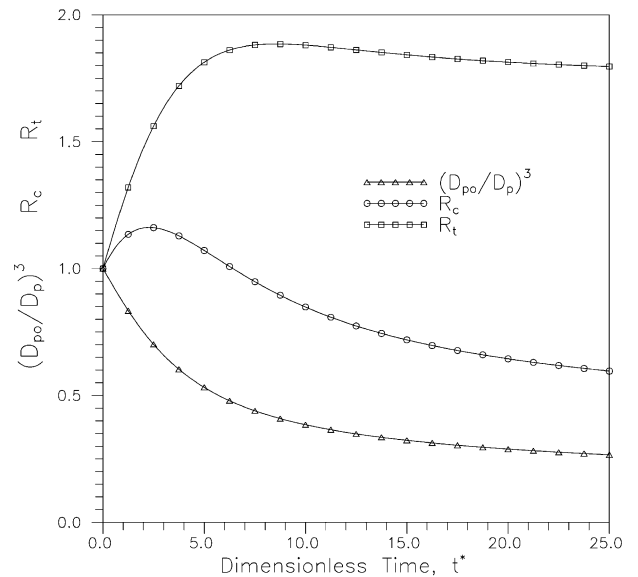


Fig. 1. Variation of $\chi = (D_{po}/D_p)^3$, $R_c = \sqrt{6E_c \chi / e^2 \Theta \rho_s \pi D_{po}^3}$ and $R_t = \left(D_{po} \left| \frac{dU_{s,x}}{dy} \right| \right) / \left(\sqrt{3} \chi^{\frac{1}{3}} \Theta^{\frac{1}{2}} \right)$ as a function of dimensionless time for shearing flow.

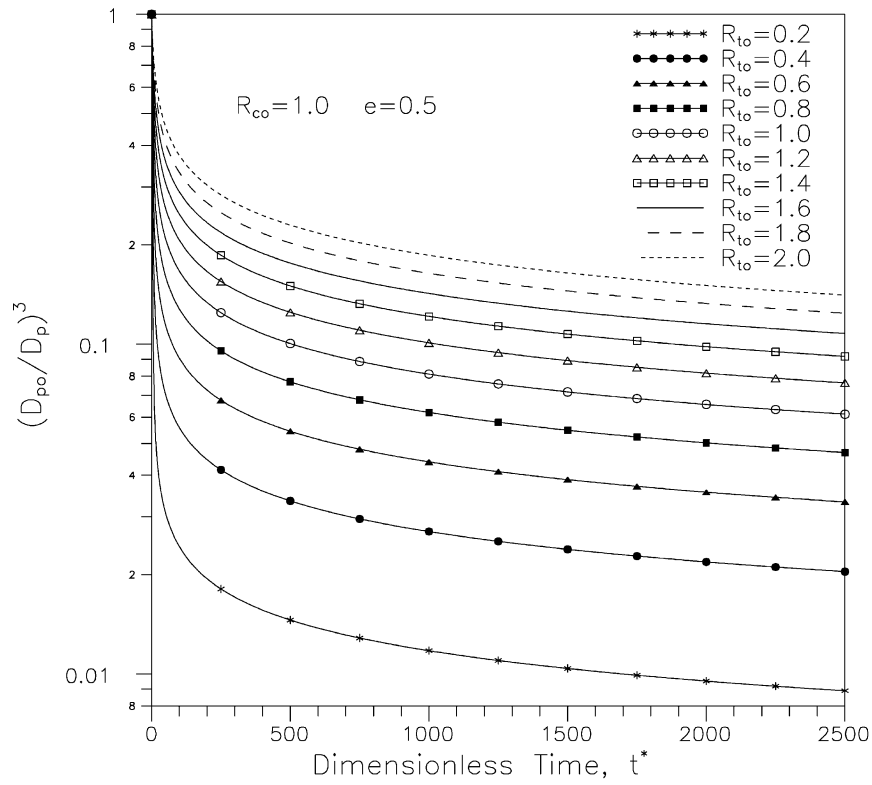


Fig. 2. Effect of shear rate on the average particle size as a function of dimensionless time.

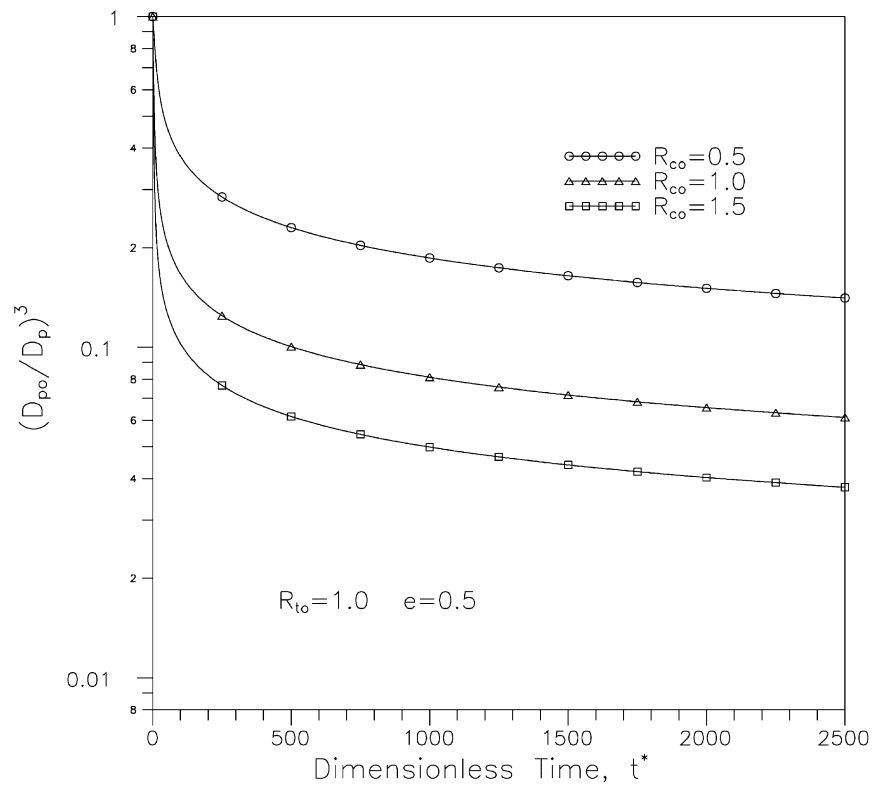


Fig. 3. Effect of cohesive surface energy on the average particle size as a function of dimensionless time.

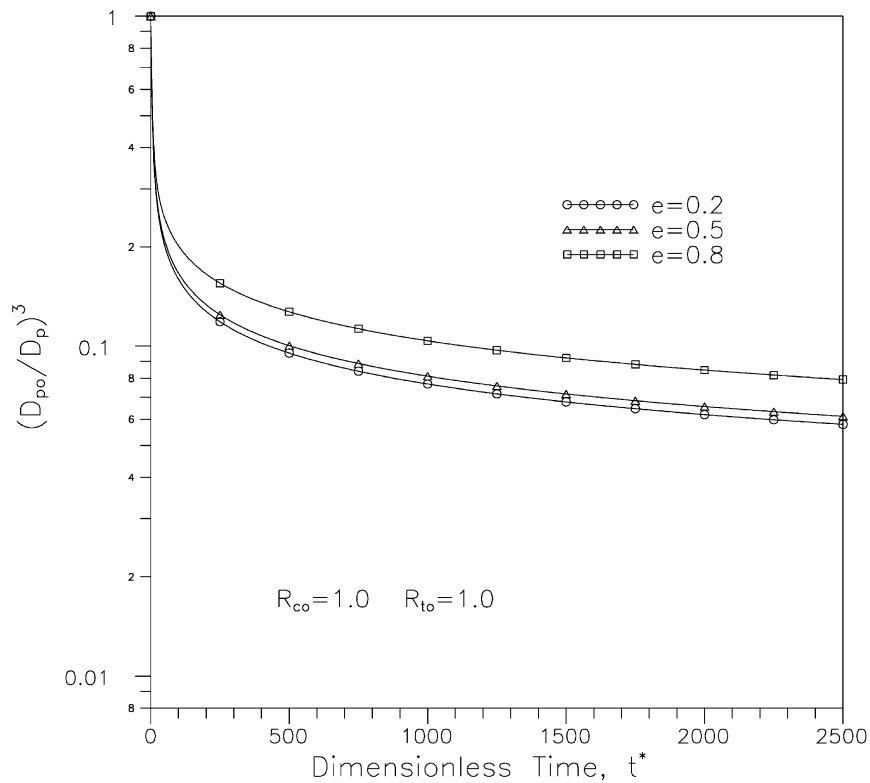


Fig. 4. Effect of restitution coefficient on the average particle size as a function of dimensionless time.

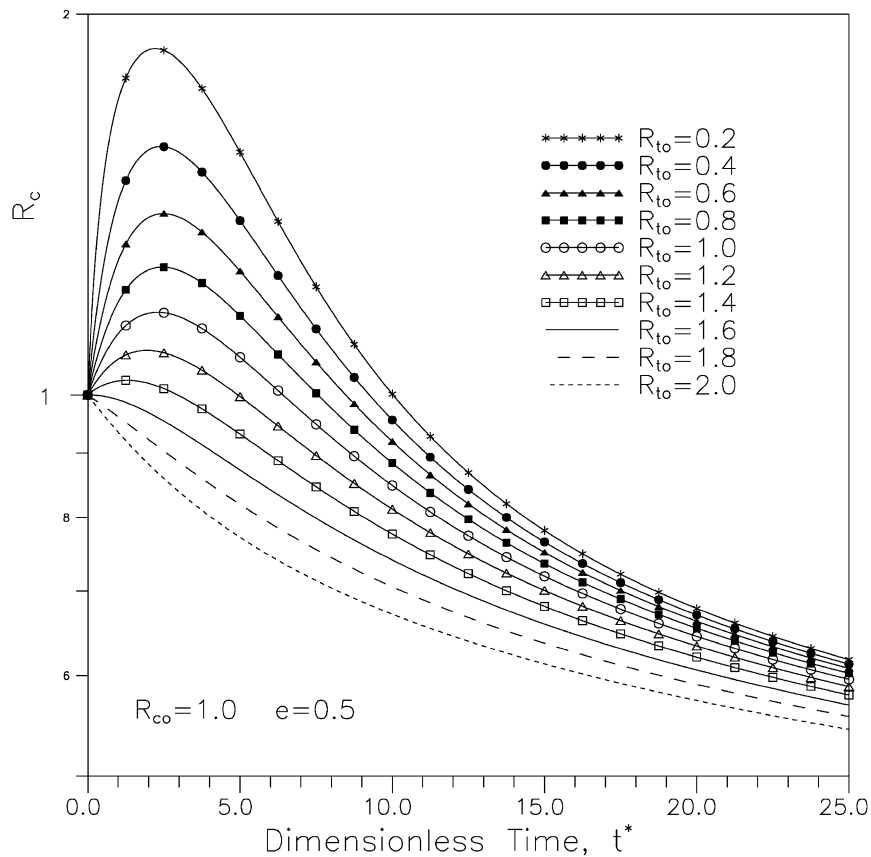


Fig. 5. Effect of shear rate on R_c as a function of dimensionless time.

ratio of initial particle diameter to average particle diameter (χ) sharply decreases. The ratio of shear velocity to fluctuation velocity (R_t) sharply increases, and the ratio of the contacting bonding energy to the fluctuation energy (R_c) slightly increases then decreases. This is anticipated since agglomeration results first in the decrease of fluctuation velocity followed by a decrease in contacting bonding energy. An initial increase of the average particle size is due to significant surface (e.g., electrostatic) forces of the smaller particles. As the average particle/agglomerate size increases, the contacting bonding energy decreases due to significantly lower surface forces between bigger particles or agglomerates. On the other hand, the fluctuation energy dissipation increases due to the initial agglomeration, then decreases later due to more chaotic behavior of less cohesive particles/agglomerates. This, in turn, results in higher viscous kinetic energy. After a long period of time, particle size reaches its maximum value, with no further agglomeration. Thus, the particle size, R_c , and R_t approach their steady state values. Although our model does not impose the deagglomeration, the terminal particle size is achieved due to the lower collisional frequency and lower surface forces between big particles/agglomerates. Fig. 2 shows the behavior of particle diameter ratio (χ) as a function of time at various R_{t0} , different shear rates, and constant initial fluctuation velocity. The overall trends of the parti-

cle diameter variation did not change with the shear rate; however, higher shear rate (higher values of R_{t0}) resulted in smaller particle/agglomerate diameter. Fig. 3 shows the behavior of particle diameter ratio (χ) as a function of time at various R_{co} , different contact bonding energy (cohesive surface energy) and constant initial fluctuation energy. The overall trends of particle diameter variation did not change with an increase in the cohesive surface energy of the particle; however, as expected, larger particles/agglomerates were obtained at higher cohesive surface energy (higher values of R_{co}). Fig. 4 shows the behavior of particle diameter ratio (χ) as a function of time at various e , restitution coefficients. The overall trends in particle diameter variation did not change with an increase in the restitution coefficient; however, as was expected, a higher restitution coefficient resulted in a slightly smaller particle/agglomerate diameter. Fig. 5 shows the short-time behavior of the ratio of the contacting bonding energy to the fluctuation energy (R_c), and Fig. 6 shows the short-time behavior of the ratio of shear velocity to fluctuation velocity (R_t), at different shear rates and constant initial fluctuation velocity. Initially at lower R_{t0} , shear is not sufficient to initiate particle fluctuation; therefore, particles have more tendency to agglomerate. This resulted in the initial increase of both R_c and R_t at lower R_{t0} due to significantly higher energy dissipation caused

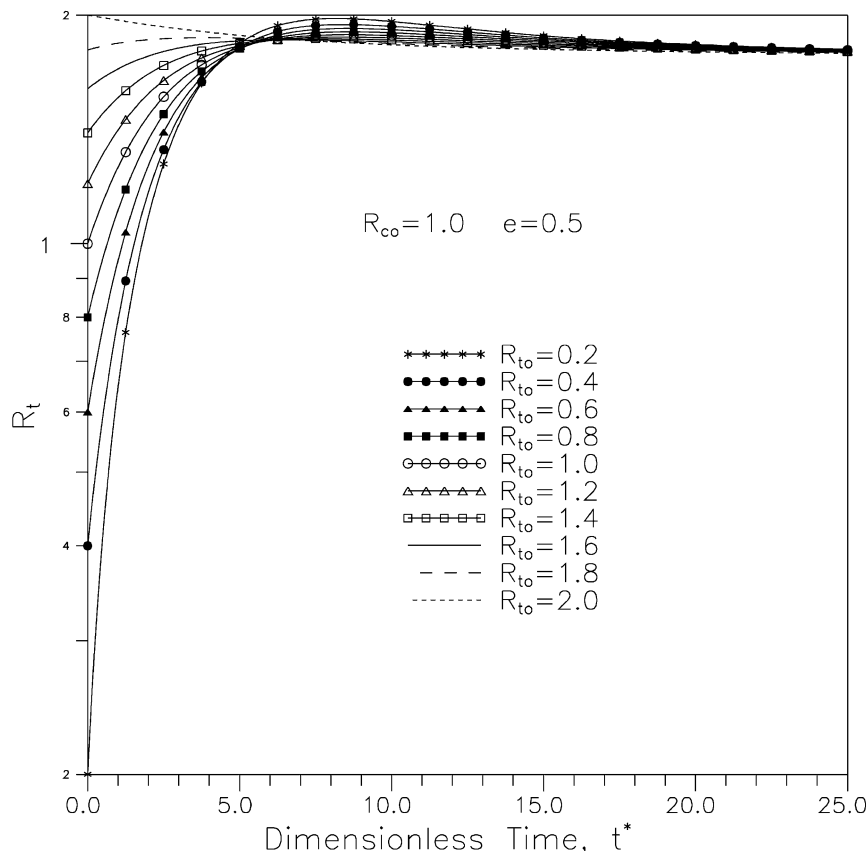


Fig. 6. Effect of shear rate on R_t as a function of dimensionless time.

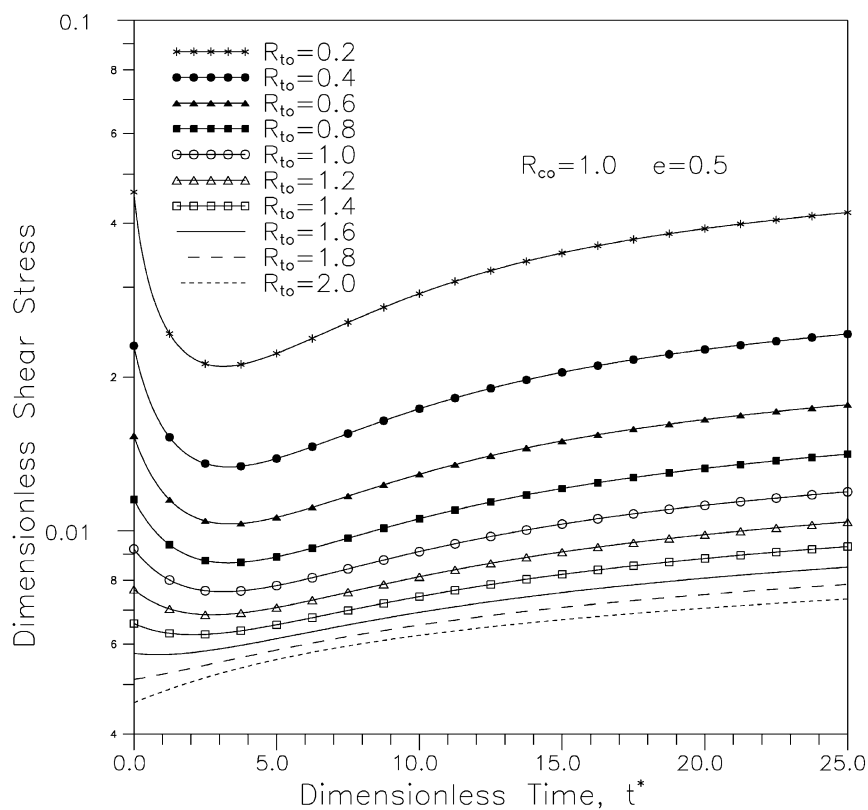


Fig. 7. Effect of shear rate on dimensionless shear stress as a function of dimensionless time.

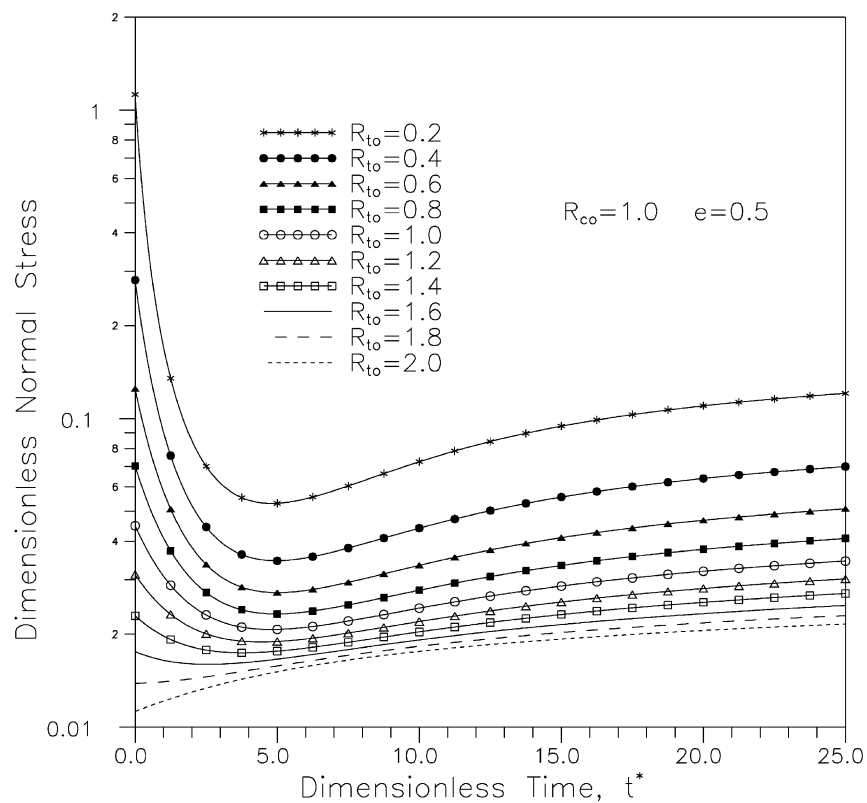


Fig. 8. Effect of shear rate on dimensionless normal stress as a function of dimensionless time.

by particle agglomerations and/or collisions of larger particles/agglomerates. Later on as agglomeration slows down due to the presence of larger particles, or at higher R_{10} when shear is sufficient to initiate significant particle fluctuation, the collision of non-agglomerating particles became dominant; therefore, both R_c and R_t decreased due to higher viscous force and particle fluctuation. Fig. 7 shows the short-time behavior of dimensionless shear stress and Fig. 8 shows the short-time behavior of dimensionless normal stress at different shear rates and constant initial fluctuation velocity. Our calculation showed that when the shear rate is low, initially the particle agglomeration is dominant and flow becomes less chaotic, so both shear and normal stresses decrease. Later on, collision of non-agglomerating particles becomes dominant and results in enhancement of fluctuation velocity and, in turn, both shear and normal stresses increase. On the other hand, as the shear rate increases, collision of particles becomes dominant; thus, both shear and normal stresses keep increasing due to enhancement of fluctuation velocity. Figs. 7 and 8 clearly show this behavior. Overall, shear rates and surface forces as a result of cohesiveness and different restitution coefficient of particles affect particle collisions and, in turn, the extent of agglomeration as well as the particulate phase flow parameters such as shear and normal stresses.

6. Conclusions

(1) Mathematical equations were developed for cohesive particle collision based on the criterion of agglomeration in order to account for the effect of cohesiveness of particles and particle agglomeration during collision. This resulted in an effective restitution coefficient, which is a function of contact bonding energy and relative velocity of particles before collision.

(2) Kinetic theory for the particulate phase was modified. The modification included the development of the distribution function of the particle instantaneous velocity based on the solid volume fraction. This enabled us to describe the particle size variation due to the formation of agglomerates without evaluating the Maxwell–Boltzmann integral–differential equation at each agglomeration occurrence.

(3) The governing equations for flow of cohesive particles were derived: the continuity, momentum, and energy equations for the particulate phase, and the conservation of the number of particles equation which accounts for particle size variation upon agglomeration.

(4) Particulate phase flow parameters were significantly affected by the shear rates, the surface forces as a result of the cohesiveness, and/or the inelasticity of the particles.

(5) The predicted flow properties such as dimensionless shear stress, dimensionless normal stress, and particle

growth agreed well with the expected trend for homogeneous shear flow.

(6) More accurate and experimentally verified agglomeration and attrition formulation is needed to refine our model predictions such as: size reduction criteria upon collision, and more rigorous mathematical integration over the agglomerating domain.

Nomenclature

$\vec{c}$	Instantaneous velocity of particle before collision
$\vec{C}$	Peculiar velocity of particle before collision = $\vec{c} - \vec{U}_s$
$\vec{c}'$	Instantaneous velocity of particle after collision
D_p	Particle diameter
D_{po}	Initial particle diameter before agglomeration
e	Normal restitution coefficient
E_c	Contact bonding energy, g $\mu\text{m}^2/\text{s}^2$
ΔE	Energy loss due to collision, g $\mu\text{m}^2/\text{s}^2$
f	Tangential roughness coefficient
g_o	Radial distribution function
k_s	Conductivity of fluctuation energy, kg/m s
$\vec{k}$	Unit vector pointing from particle 1 to particle 2
m	Mass of particle
n	Number of particles per unit volume
N_a	Number of agglomerations per unit volume and time
p_s^c	Collisional solid phase pressure
p_s^k	Kinetic solid phase pressure
$\vec{q}_c$	Flux of fluctuation energy, kg/s ³
R_c	Ratio of contacting bonding energy to fluctuation energy
R_t	Ratio of mean shear velocity to r.m.s. of instantaneous fluctuation velocity
$\vec{r}$	Position vector of particle
$\mathbf{S}$	Shear stress tensor, kg/m s ²
t	Time, s
t^*	Dimensionless time
$\mathbf{T}_f$	Frictional stress, kg/m s ²
$\mathbf{T}_s$	Solid phase stress tensor, kg/ms ²
$\vec{U}_s$	Mean velocity of solid phase, m/s
V_p	Volume of particle
V_{po}	Initial volume of particle
V_c	Agglomeration limit of instantaneous velocity difference
γ	Inter-particle collisional energy dissipation, kg/m s ³
ε_s	Volume fraction of solid phase
ε_m	Maximum packing volume fraction of solid phase
Θ	Fluctuation energy of particulate phase, m ² /s ²
μ_b	Bulk viscosity, kg/m s
μ_f	Frictional viscosity, kg/m s
μ_s	Shear viscosity, kg/m s
ξ_{a1}, ξ_{a2}	Agglomeration number coefficient
ξ_{p1}, c_{p2}	Cohesive pressure and viscosity coefficient
$\xi_{\gamma1}, \xi_{\gamma2}$	Cohesive dissipation coefficient

ρ_s	Density of particle, kg/m ³
χ	Ratio of initial volume of particle to volume of agglomerate

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